

Brain Booster – An Online Learning and Assessment Platform

Abstract:

In the modern educational era, digital learning platforms play a significant role in providing accessible and structured learning experiences for students. However, many platforms lack an integrated system for performance evaluation before enrollment, resulting in mismatched course selection and poor learning outcomes. **Brain Booster** is a comprehensive web application designed to bridge this gap by offering a structured learning path that begins with an **initial assessment test**, ensuring that only students with a minimum skill level can enroll in advanced courses.

The platform involves three main actors: **Admin**, **Mentor**, and **Student**. Mentors can create and upload courses, manage bookings, and track student progress. Students begin their journey by taking an initial test, and only those who achieve a score above **50%** are eligible to enroll in courses. Once enrolled, students gain access to a dedicated **learning area**, where they can access course content, assignments, and resources.

The system also features **exam modules**, enabling students to take exams and receive instant results with detailed performance analytics. Additionally, the platform integrates with external resources, allowing students to access **digital textbooks** and other study materials, making it a one-stop solution for academic growth.

Admins play a crucial role in maintaining platform integrity by verifying student and mentor accounts, monitoring activity, and ensuring quality control. With its structured approach and easy-to-use interface, **Brain Booster** provides an effective platform for improving student learning outcomes and mentor-student collaboration.

Actors:

1. **Admin** – Manages verification of mentors and students, monitors courses and activities, and ensures platform quality.
 2. **Mentor** – Creates and manages courses, uploads content, and tracks student enrollments.
 3. **Student** – Takes initial tests, enrolls in eligible courses, attends exams, and accesses study materials.
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Modules:

1. Course Management (Mentor Panel)

- Mentors can **create courses** with video lectures, notes, and assessments.
- Manage student bookings and track enrollment trends.
- Update or delete courses as needed.

2. Admin Verification and Monitoring

- Verify new student and mentor accounts before activation.
- Monitor ongoing activities, courses, and exams to ensure quality standards.
- Manage reports of inappropriate content or misconduct.

3. Student Learning and Enrollment Module

- **Initial Test System:**
 - Students must take a test before enrolling in any course.
 - A minimum of **50% score** is required to unlock course enrollment.
- After enrolling, students gain access to a **learning area** with interactive modules.
- Students can take **exams** related to their courses and view their marks instantly.

4. Exam and Performance Tracking

- Integrated exam system with automated evaluation.
- Instant grading and performance analytics provided to students.
- Mentors can track the performance of their students through reports.

5. External Resource Integration

- A dedicated **resource area** where students can access **digital textbooks**, references, and additional learning materials.
- Resources can be integrated via external APIs or manual uploads.

Tech Stack:

- **Frontend:** React.js, Styled-Components / TailwindCSS
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **External Integration:** Google Books API (or similar) for textbook integration
- **Hosting & Deployment:** AWS / Vercel / Render

Software Requirements:

Hardware:

- **Processor:** Intel i3 or higher
- **RAM:** 4GB minimum
- **Storage:** 250GB HDD/SSD

Software:

- **Operating System:** Windows / Linux / MacOS
 - **Development Tools:** VS Code, Postman, Git
 - **Web Technologies:** React.js, Node.js, MongoDB
 - **API Services:** External API for textbook integration
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Key Features:

- Initial test-based enrollment to ensure students are ready for advanced courses.
 - Mentor-driven course creation and management system.
 - Real-time exam and instant evaluation system.
 - Dedicated learning area for enrolled courses.
 - External textbook integration for enhanced study support.
 - Secure admin verification to maintain platform quality.
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Workflow:

1. **Student Registration → Initial Test → Eligibility Check → Course Enrollment → Learning & Exams → Certification (Optional).**
 2. Mentors create and manage courses while tracking bookings.
 3. Admin verifies accounts and maintains platform security.
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Conclusion:

Brain Booster provides a structured and performance-driven learning platform that ensures students are adequately prepared before enrolling in courses. By integrating features like initial testing, course management, external resources, and real-time exams, the platform creates a holistic environment for learning and growth. It not only benefits students by offering quality education but also empowers mentors to manage and monitor their courses effectively.